positions and velocities whether there were ice cubes in the glass would be a Herculean task. So we turned instead to probabilistic reasoning, which is computationally tractable and, moreover, deals with the macroscopic properties— order versus disorder; for example, ice versus water—we are generally interested in. But keep in mind that probability is by no means fundamentally stitched into the fabric of classical physics. In principle, if we knew precisely how things were now—knew the positions and velocities of every single particle making up the universe—classical physics says we could use that information to predict how things would be at any given moment in the future or how they were at any given moment in the past. Whether or not you actually follow its moment-to-moment development, according to classical physics you can talk about the past and the future, in principle, with a confidence that is controlled by the detail and the accuracy of your observations of the present.[1]

Probability will also play a central role in this chapter. But because probability *is* an inescapable element of quantum mechanics, it fundamentally alters our conceptualization of past and future. We've already seen that quantum uncertainty prevents simultaneous knowledge of exact positions and exact velocities. Correspondingly, we've also seen that quantum physics predicts only the probability that one or another future will be realized. We have confidence in these probabilities, to be sure, but since they are probabilities we learn that there is an *unavoidable* element of chance when it comes to predicting the future.

When it comes to describing the past, there is also a critical difference between classical and quantum physics. In classical physics, in keeping with its egalitarian treatment of all moments in time, the events leading up to something we observe are described using exactly the same language, employing exactly the same attributes, we use to describe the observation itself. If we see a fiery meteor in the night sky, we talk of its position and its velocity; if we reconstruct how it got there, we also talk of a unique succession of positions and velocities as the meteor hurtled through space toward earth. In quantum physics, though, once we observe something we enter the rarefied realm in which we know something with 100 percent certainty (ignoring issues associated with the accuracy of our equipment, and the like). But the past—by which we specifically mean the "unobserved" past, the time before we, or anyone,